

RAG-Based Customer Support Assistant – Technical Documentation

1. Introduction

The RAG (Retrieval-Augmented Generation) based Customer Support Assistant combines document retrieval with LLM-powered response generation to deliver accurate, context-aware answers from a provided knowledge base. The system improves response reliability by grounding outputs in retrieved document content.

2. System Architecture

The solution consists of four core layers:

- **Ingestion Layer** – Loads and processes source documents (PDFs).
- **Embedding Layer** – Converts document chunks into vector embeddings.
- **Retrieval Layer** – Fetches relevant context using semantic similarity search.
- **Generation Layer** – Uses an LLM to generate responses based on retrieved context.

3. Design Decisions

- **Optimized chunk size** for balancing context and retrieval quality.
- **HuggingFace embeddings** for semantic document representation.
- **ChromaDB** selected as the vector store for efficient retrieval.
- **Modular architecture** for scalability and maintainability.

4. Workflow

User Query → Retrieve Relevant Context → Generate Response → Return Answer

5. Conditional Logic

- **Respond** when relevant context is available.
- **Request clarification** for vague or incomplete queries.
- **Return fallback responses** when no supporting information is found.

6. Human-in-the-Loop (HITL)

Supports human intervention for handling uncertain responses, validating outputs, and improving reliability in edge cases.

7. Challenges

- **Dependency and environment setup issues.**
- **Improving retrieval relevance and answer accuracy.**
- **Tuning chunk size for optimal performance.**

8. Testing

- **Validated system behaviour using sample user queries.**
- **Performed response verification against PDF source content.**
- **Tested retrieval and fallback scenarios.**

9. Future Enhancements

- **Support for multiple document sources.**
- **Conversational memory for multi-turn interactions.**
- **Deployment as a production-ready application.**